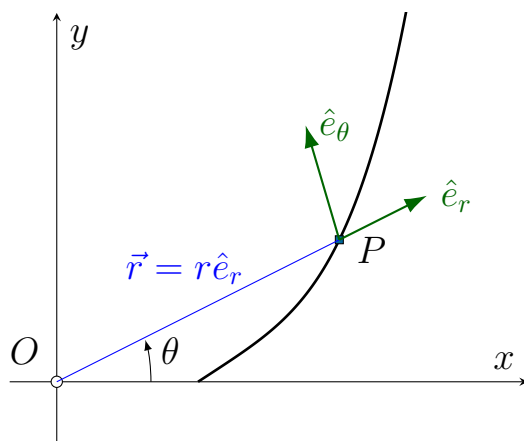


Lesson 5: 2D motion with polar coordinates

1 The Equations of Motion with Polar Coordinates



In polar coordinates we attach the unit vectors to the particle. The unit vector $\hat{e}_r$ is directed in the radial direction. The unit vector $\hat{e}_\theta$ points in the transverse direction—the direction in which the particle would move if the angle θ increases while the length of the radial vector $\vec{r}$ remains fixed.

In terms of these components, the velocity vector can be expressed as,

$$\vec{v}(t) = \dot{r}\hat{e}_r + r\dot{\theta}\hat{e}_\theta. \quad (1)$$

If we express the velocity in terms of its scalar components,

$$\vec{v} = v_r\hat{e}_r + v_\theta\hat{e}_\theta,$$

we have,

$$v_r = \dot{r}, \quad v_\theta = r\dot{\theta}.$$

The acceleration in polar coordinates is expressed as

$$\vec{a}(t) = \left(\ddot{r} - r\dot{\theta}^2\right)\hat{e}_r + \left(r\ddot{\theta} + 2\dot{r}\dot{\theta}\right)\hat{e}_\theta \quad (2)$$

If we express the acceleration in terms of its scalar components,

$$\vec{a} = a_r\hat{e}_r + a_\theta\hat{e}_\theta,$$

we have

$$a_r = \ddot{r} - r\dot{\theta}^2, \quad a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta}.$$

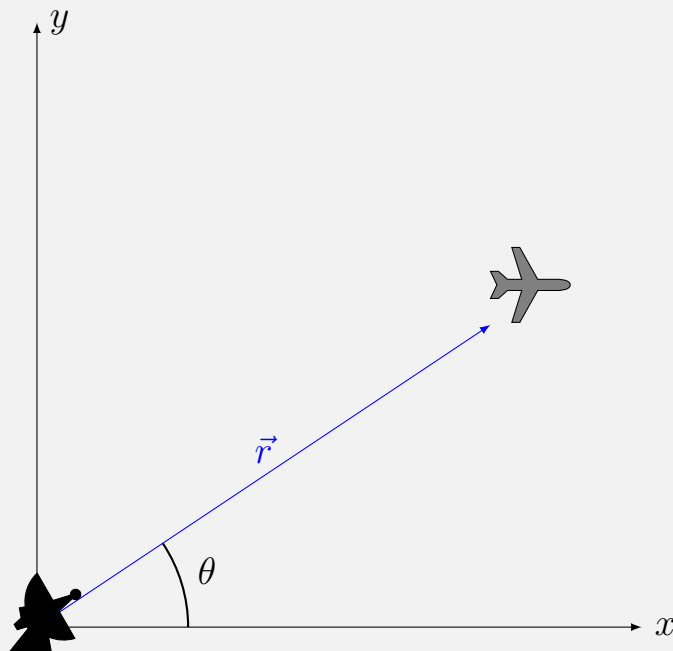
Sample Problem 5.1

A radar tracking station provides the following raw data at a given instant in time.

5.1 – cont

$$\begin{array}{ll} r = 8000 \text{ ft} & \theta = 33^\circ \\ \dot{r} = 200 \text{ ft/s} & \dot{\theta} = -0.05 \text{ rad/s} \\ \ddot{r} = 6 \text{ ft/s}^2 & \ddot{\theta} = 0.002 \text{ rad/s}^2 \end{array}$$

Based on this data, what is the instantaneous velocity and acceleration in the r and θ directions? What is the instantaneous velocity and acceleration in the x and y directions?



5.1 – cont

$$v_r = \dot{r} = 200$$

$$v_\theta = r\dot{\theta} = 8000(-0.05) = -400$$

$$\vec{v} = 200\hat{e}_r - 400\hat{e}_\theta \text{ ft/s}$$

$$v_x = 200 \cos(33) + 400 \cos(57) = 385.6$$

$$v_y = 200 \sin(33) - 400 \sin(57) = -226.5$$

$$\vec{v} = 385.6\hat{i} - 226.5\hat{j} \text{ ft/s}$$

$$a_r = \ddot{r} - r\dot{\theta}^2 = 8 - 8000(-0.05)^2 = -14 \text{ ft/s}^2$$

$$a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta} = 8000(.0002) + 2(200)(-0.05) = -4 \text{ ft/s}^2$$

$$\vec{a} = -14\hat{e}_r - 4\hat{e}_\theta \text{ ft/s}^2$$

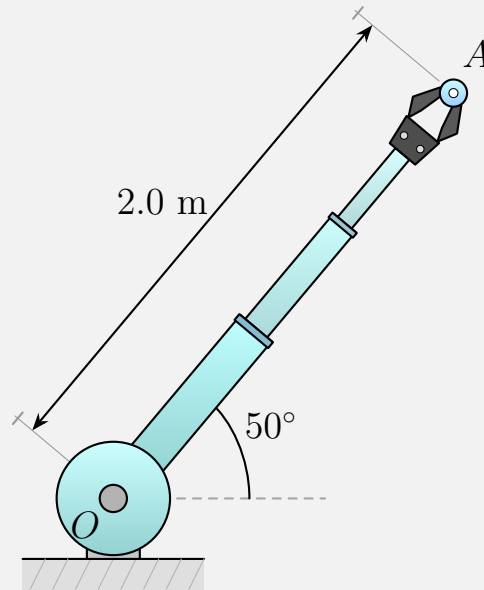
$$a_x = -14 \cos(33) - 4 \cos(57) = -9.56$$

$$a_y = -14 \sin(33) - 4 \sin(57) = -10.98$$

$$\vec{a} = -9.56\hat{i} - 10.98\hat{j} \text{ ft/s}^2$$

Sample Problem 5.2

At the instant shown, the length of the robotic arm is decreasing at a constant rate of 0.1 m/s, while being lowered by a constant rate of 0.5 rad/s. Determine (a) the magnitude of the velocity of point A and (b) the magnitude of the acceleration of point A .



From the picture we have $r = 2$ m and $\theta = 50^\circ$. From the statement of the problem we have $\dot{r} = -0.1$ m/s and $\dot{\theta} = -0.5$ rad/s. Note the minus signs in both. Since both are lowered at constant rates we have $\ddot{r} = \ddot{\theta} = 0$.

$$v_r = \dot{r} = -0.1$$

$$v_\theta = r\dot{\theta} = 2(-0.5) = -1$$

5.2 – cont

The magnitude is

$$|\vec{v}| = \sqrt{v_r^2 + v_\theta^2} = \sqrt{(-0.1)^2 + (-1)^2} = 1.005 \text{ m/s}$$

$$a_r = \ddot{r} - r\dot{\theta}^2 = 0 - 2(-0.5)^2 = -0.5 \text{ m/s}^2$$

$$a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta} = 0 + 2(-0.1)(-0.5) = 0.1 \text{ m/s}^2$$

$$\vec{a} = -0.5\hat{e}_r - 0.1\hat{e}_\theta \text{ m/s}^2$$

The magnitude is

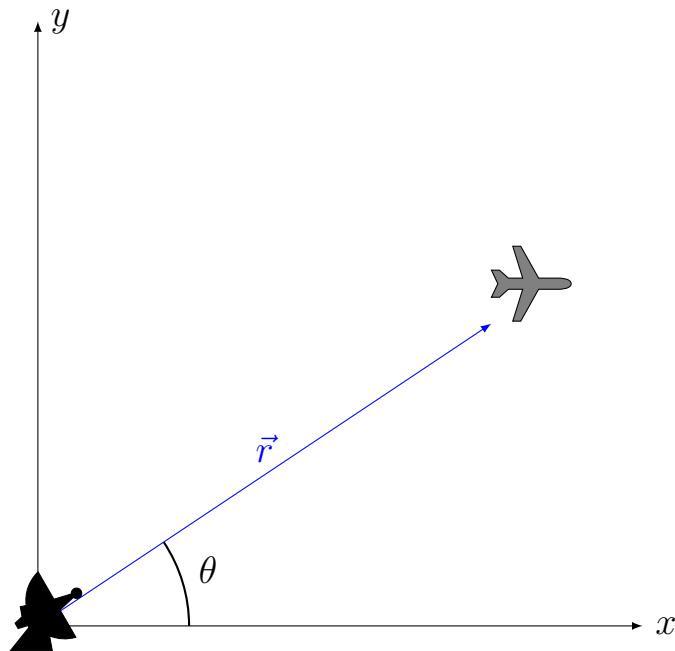
$$|\vec{a}| = \sqrt{a_r^2 + a_\theta^2} = \sqrt{(-0.5)^2 + (0.1)^2} = 0.51 \text{ m/s}$$

Sample Problems

1. A radar tracking station provides the following raw data at a given instant in time.

$$\begin{array}{ll} r = 8000 \text{ ft} & \theta = 33^\circ \\ \dot{r} = 200 \text{ ft/s} & \dot{\theta} = -0.05 \text{ rad/s} \\ \ddot{r} = 6 \text{ ft/s}^2 & \ddot{\theta} = 0.002 \text{ rad/s}^2 \end{array}$$

Based on this data, what is the instantaneous velocity and acceleration in the r and θ directions? What is the instantaneous velocity and acceleration in the x and y directions?



2. At the instant shown, the length of the robotic arm is decreasing at a constant rate of 0.1 m/s , while being lowered by a constant rate of 0.5 rad/s . Determine (a) the magnitude of the velocity of point A and (b) the magnitude of the acceleration of point A .

